STAT 526: HW 6

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Problem 0

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Problem 1

> Researchers study the effect of seat belt use on the severity of traffic accidents. The counts of events are summarized below.

a.

> Describe the sampling process that would result in this being a retrospective study.

Researchers collect data on individuals who were in traffic accidents and then we label them with fatal vs non-fatal and seat belt and no seat belt.

b.

> Describe the sampling process that would result in a this being a prospective study.

We select two groups before accidents occur. After accidents, we record fatal vs non-fatal and seat belt and no seat belt.

c.

For odds ratio (OR):

$$\widehat{OR} = \frac{n_{11} \times n_{22}}{n_{12} \times n_{21}} = \frac{1061 \times 421,368}{501 \times 165,227} \approx 5.401$$

$$SE(\ln \widehat{OR}) = \sqrt{\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}} = \sqrt{\frac{1}{1061} + \frac{1}{501} + \frac{1}{165,227} + \frac{1}{421,368}} \approx 0.0543$$

The CI for $\ln(OR) = \ln(5.401) = 1.687$ is $1.687 \pm 1.96(0.0543) = [1.581, 1.793]$. Exponentiating the bounds gives the 95% CI for OR: [4.86, 6.01].

With 95% confidence, the the odds of a fatal accident for those not wearing a seat belt are between 4.86 and 6.01 times higher than the odds for those wearing a seat belt.

Difference in Proportions

$\hat{p}_1 = \frac{1061}{1061+501} \approx 0.6793$ and $\hat{p}_2 = \frac{165{,}227}{165{,}227+421{,}368} \approx 0.2817$ so we can find:

$$\text{Difference} = \hat{p}_1 - \hat{p}_2 = 0.6793 - 0.2817 = 0.3976$$

Now for SE (of binomial):

$$SE = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} = \sqrt{\frac{0.6793(0.3207)}{1562} + \frac{0.2817(0.7183)}{586,595}} \approx 0.0118$$

Finally, all together: 95% CI is $0.3976 \pm 1.96(0.0118) = [0.374, 0.421]$

Now we are 95% confident that the probability of having a fatal accident is increased by between 37.4 and 42.1 percent higher for individuals not wearing their seat belts versus those who were wearing their seat belts.

Problem 2 - Faraway Chapter 6 Exercise #4

The dataset death contains data on murder cases in Florida in 1977. The data is cross-classified by the race (black or white) of the victim, of the defendant and whether the death penalty was given.

a.)

Consider the frequency with which the death penalty is applied to black and white defendants, both marginally and conditionally, with respect to the race of the victim. Is this an example of Simpson's paradox? Are the observed differences in the frequency of application of the death penalty statistically significant?

```r
library(faraway)
data(death)
print(death)
```

```
##     y penalty victim defend
## 1  19     yes      w      w
## 2 132      no      w      w
## 3   0     yes      b      w
## 4   9      no      b      w
## 5  11     yes      w      b
## 6  52      no      w      b
## 7   6     yes      b      b
## 8  97      no      b      b
```

Calculate frequencies:

Marginal (just looking without race)

```r
marginal_df <- xtabs(y ~ defend + penalty, data=death)
prop.table(marginal_df, 1)
```

```
##       penalty
## defend        no       yes
##      b 0.8975904 0.1024096
##      w 0.8812500 0.1187500
```

```r
# Simple Chi-square test
chisq.test(marginal_df)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  marginal_df
## X-squared = 0.086343, df = 1, p-value = 0.7689
```

Marginally, white gets more death penalty. This is not statistically significant.

Now conditioned on victim race:

```r
ftable(xtabs(y ~ victim + defend + penalty, data=death))
```

```
##                penalty  no yes
## victim defend
## b      b               97   6
##        w                9   0
## w      b               52  11
##        w              132  19
```

```r
conditional_tab <- xtabs(y ~ defend + penalty + victim, data=death)
prop.table(conditional_tab, c(1, 3))
```

```
## , , victim = b
##
##       penalty
## defend         no        yes
##      b 0.94174757 0.05825243
##      w 1.00000000 0.00000000
##
## , , victim = w
##
##       penalty
## defend         no        yes
##      b 0.82539683 0.17460317
##      w 0.87417219 0.12582781
```

For both races, black people are more likely to get death penalty.

YES, this is example of Simpson's paradox. The paradox is that the trend appears when only looking at it marginally, but when we look at the conditional trend, black defendants have higher death penalty rate.

Ok let us look at statistical significance

```r
cond_white_victim <- conditional_tab[,, "w"]
cond_black_victim <- conditional_tab[,, "b"]

print("Conditional Table (Victim = White):")
```

```
## [1] "Conditional Table (Victim = White):"
```

```r
print(cond_white_victim)
```

```
##       penalty
## defend  no yes
##      b  52  11
##      w 132  19
```

```r
print("Proportions (Victim = White):")
```

```
## [1] "Proportions (Victim = White):"
```

```r
print(prop.table(cond_white_victim, 1))
```

```
##       penalty
## defend        no       yes
##      b 0.8253968 0.1746032
##      w 0.8741722 0.1258278
```

```r
print("Conditional Table (Victim = Black):")
```

```
## [1] "Conditional Table (Victim = Black):"
```

```r
print(cond_black_victim)
```

```
##       penalty
## defend no yes
##      b 97   6
##      w  9   0
```

```r
print("Proportions (Victim = Black):")
```

```
## [1] "Proportions (Victim = Black):"
```

```r
print(prop.table(cond_black_victim, 1))
```

```
##       penalty
## defend         no        yes
##      b 0.94174757 0.05825243
##      w 1.00000000 0.00000000
```

```r
# Chi-squared tests for Conditional Independence
chisq.test(cond_white_victim, correct = FALSE)
```

```
##
##  Pearson's Chi-squared test
##
## data:  cond_white_victim
## X-squared = 0.87739, df = 1, p-value = 0.3489
```

```r
chisq.test(cond_black_victim, correct = FALSE)
```

```
## Warning in chisq.test(cond_black_victim, correct = FALSE): Chi-squared
## approximation may be incorrect
```

```
##
##  Pearson's Chi-squared test
##
## data:  cond_black_victim
## X-squared = 0.55395, df = 1, p-value = 0.4567
```

```r
# Cochran-Mantel-Haenszel test (overall conditional significance, doesnt really provide much)
mantelhaen.test(conditional_tab)
```

```
##
##  Mantel-Haenszel chi-squared test with continuity correction
##
## data:  conditional_tab
## Mantel-Haenszel X-squared = 0.79633, df = 1, p-value = 0.3722
## alternative hypothesis: true common odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.2864136 1.4091583
## sample estimates:
## common odds ratio
##         0.6352968
```

Our two chi-square tests for conditional independence show no significant difference in the application of death penalty between races for the case the victim is white and for the case where the victim is black.

b.)

> Determine the most appropriate dependence model between the variables.

We have 3 binary variables, victim race, defendant race, and whether there is death penalty. We do see a conditional relationship based on whether the defendant is white or black have different death penalties rates conditional on victim. More formally, whether penalty is conditionally independent of defendant's race given the victim's race is what we want to test for.

```r
# Model 1: Only Defendant's Race (Marginal)
model_marg <- glm(penalty ~ defend, family = binomial, weights = y, data = death)
summary(model_marg)
```

```
##
## Call:
## glm(formula = penalty ~ defend, family = binomial, data = death,
##     weights = y)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.1707     0.2560   -8.48   <2e-16 ***
## defendw       0.1664     0.3539    0.47    0.638
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 226.51  on 6  degrees of freedom
## Residual deviance: 226.29  on 5  degrees of freedom
## AIC: 230.29
##
## Number of Fisher Scoring iterations: 5
```

```r
# Model 2: Adding Victim's Race (Condition on defendant and victim)
model_adj <- glm(penalty ~ defend + victim, family = binomial, weights = y, data = death)
summary(model_adj)
```

```
##
## Call:
## glm(formula = penalty ~ defend + victim, family = binomial, data = death,
##     weights = y)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.8421     0.4203  -6.762 1.37e-11 ***
## defendw      -0.4402     0.4009  -1.098   0.2722
## victimw       1.3242     0.5193   2.550   0.0108 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 226.51  on 6  degrees of freedom
## Residual deviance: 219.08  on 4  degrees of freedom
## AIC: 225.08
##
## Number of Fisher Scoring iterations: 5
```

```r
# Model 3: Adding an Interaction Term (Interaction between defendant and victim)
model_int <- glm(penalty ~ defend * victim, family = binomial, weights = y, data = death)
summary(model_int)
```

```
##
## Call:
## glm(formula = penalty ~ defend * victim, family = binomial, data = death,
##     weights = y)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)       -2.7830     0.4207  -6.615 3.71e-11 ***
## defendw          -14.2430  1006.6075  -0.014   0.9887
## victimw            1.2296     0.5358   2.295   0.0217 *
## defendw:victimw   13.8579  1006.6076   0.014   0.9890
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 226.51  on 6  degrees of freedom
## Residual deviance: 218.38  on 3  degrees of freedom
## AIC: 226.38
##
## Number of Fisher Scoring iterations: 14
```

```r
# Test ANOVA for nested models
anova(model_adj, model_int, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: penalty ~ defend + victim
## Model 2: penalty ~ defend * victim
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         4     219.08
## 2         3     218.38  1  0.70074   0.4025
```

The logistic regression models seem to say only victim is important in determining rate of penalty. Perhaps we can look into Poisson GLM for counts independence instead:

```r
# Conditional model victim on penalty and victim on defendant
model_cond_indep <- glm(y ~ victim*penalty + victim*defend, family = poisson, data = death)
summary(model_cond_indep)
```

```
##
## Call:
## glm(formula = y ~ victim * penalty + victim * defend, family = poisson,
##     data = death)
##
## Coefficients:
##                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)          4.5797     0.1011  45.314  < 2e-16 ***
## victimw             -0.5876     0.1639  -3.586 0.000336 ***
## penaltyyes          -2.8717     0.4196  -6.843 7.75e-12 ***
## defendw             -2.4375     0.3476  -7.013 2.34e-12 ***
## victimw:penaltyyes   1.0579     0.4635   2.282 0.022471 *
## victimw:defendw      3.3116     0.3786   8.748  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 395.9153  on 7  degrees of freedom
## Residual deviance:   1.8819  on 2  degrees of freedom
## AIC: 49.563
##
## Number of Fisher Scoring iterations: 4
```

```r
# Saturated model
model_all_2way <- glm(y ~ (victim + penalty + defend)^2, family = poisson, data = death)
summary(model_all_2way)
```

```
##
## Call:
## glm(formula = y ~ (victim + penalty + defend)^2, family = poisson,
##     data = death)
##
## Coefficients:
```

```
##                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)           4.5781     0.1012  45.230  < 2e-16 ***
## victimw              -0.6331     0.1714  -3.694 0.000221 ***
## penaltyyes           -2.8421     0.4203  -6.761 1.37e-11 ***
## defendw              -2.4177     0.3480  -6.947 3.72e-12 ***
## victimw:penaltyyes    1.3242     0.5193   2.550 0.010779 *
## victimw:defendw       3.3580     0.3820   8.791  < 2e-16 ***
## penaltyyes:defendw   -0.4402     0.4009  -1.098 0.272155
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 395.91531  on 7  degrees of freedom
## Residual deviance:   0.70074  on 1  degrees of freedom
## AIC: 50.382
##
## Number of Fisher Scoring iterations: 4
```

```r
# Compare the two models
anova(model_cond_indep, model_all_2way, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: y ~ victim * penalty + victim * defend
## Model 2: y ~ (victim + penalty + defend)^2
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         2    1.88190
## 2         1    0.70074  1   1.1812   0.2771
```

The model where defendant and penalty are conditional on victim seems to perform best. Good residual deviance values, matching chi-squared, and also statistically the saturated model is not a significant improvement. Things look good here. In fact, from the saturated model coefficients, it seems that the penalty:defender interaction is not significant.

Our final and most appropriate model is the poisson glm model that models the conditional independence using coefficients for victim, defendant, penalty, victim:defendant and victim:penalty.

c.)

> Fit a binomial regression with death penalty as the response and show the relationship to your model in the previous question.

Good thing we already did all the work. We can paste it here again:

```r
# Model 1: Only Defendant's Race (Marginal)
model_marg <- glm(penalty ~ defend, family = binomial, weights = y, data = death)
summary(model_marg)
```

```
##
## Call:
## glm(formula = penalty ~ defend, family = binomial, data = death,
```

```
##     weights = y)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.1707     0.2560   -8.48   <2e-16 ***
## defendw       0.1664     0.3539    0.47    0.638
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
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## AIC: 230.29
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```r
# Model 2: Adding Victim's Race (Condition on defendant and victim)
model_adj <- glm(penalty ~ defend + victim, family = binomial, weights = y, data = death)
summary(model_adj)
```

```
##
## Call:
## glm(formula = penalty ~ defend + victim, family = binomial, data = death,
##     weights = y)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.8421     0.4203  -6.762 1.37e-11 ***
## defendw      -0.4402     0.4009  -1.098   0.2722
## victimw       1.3242     0.5193   2.550   0.0108 *
## ---
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##
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##
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```r
# Model 3: Adding an Interaction Term (Interaction between defendant and victim)
model_int <- glm(penalty ~ victim*defend, family = binomial, weights = y, data = death)
summary(model_int)
```

```
##
## Call:
## glm(formula = penalty ~ victim * defend, family = binomial, data = death,
##     weights = y)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
```

```
## (Intercept)       -2.7830     0.4207  -6.615 3.71e-11 ***
## victimw            1.2296     0.5358   2.295   0.0217 *
## defendw          -14.2430  1006.6075  -0.014   0.9887
## victimw:defendw   13.8579  1006.6076   0.014   0.9890
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 226.51  on 6  degrees of freedom
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## AIC: 226.38
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## Number of Fisher Scoring iterations: 14
```

```r
# Test ANOVA for nested models
anova(model_adj, model_int, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: penalty ~ defend + victim
## Model 2: penalty ~ victim * defend
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         4     219.08
## 2         3     218.38  1  0.70074   0.4025
```

We set up 3 binomial models with penalty as our binomial response. Marginal model of defendant alone is not significant as a coefficient, which makes sense since we think it is conditional independence. We see that in the more complex models, whether the victim is white or not is significant. Although, victim:defendant does not seem to be the case.

To see the relationship, we are looking at the last model model_int. It tells us that the death penalty rate conditioned on victim is independent (insignificant p-value) from defendant on victim. This is the conditional independence model that we saw in the Poisson GLM models.

Problem 3

A recent poll looked at the relationships between three categorical variables (State (1-5), county type (urban/rural), and whether the respondent supports Trump's reelection (yes/no)).

a.)

Test the independence of county type and Trump support, unconditionally on state, by means of an odds ratio.

Ok we can first ignore state then and compute the odds ratio.

```r
poll_data <- data.frame(
  State = factor(rep(1:5, each = 4)),
  County = factor(rep(c("Urban", "Rural", "Urban", "Rural"), 5)),
  Support = factor(rep(c("Yes", "Yes", "No", "No"), 5)),
  Freq = c(
    53, 17, 115, 15, # State 1
    71, 33, 75, 21,  # State 2
    58, 40, 81, 21,  # State 3
    34, 34, 111, 21, # State 4
    71, 42, 75, 12   # State 5
  )
)

marginal_table <- xtabs(Freq ~ County + Support, data = poll_data)
print(marginal_table)
```

```
##        Support
## County   No Yes
##   Rural  90 166
##   Urban 457 287
```

```r
odds_ratio <- (marginal_table["Urban", "Yes"] * marginal_table["Rural", "No"]) /
               (marginal_table["Urban", "No"] * marginal_table["Rural", "Yes"])
print(odds_ratio)
```

```
## [1] 0.3404867
```

```r
chisq.test(marginal_table, correct = FALSE)
```

```
##
##  Pearson's Chi-squared test
##
## data:  marginal_table
## X-squared = 53.039, df = 1, p-value = 3.27e-13
```

Or we can calculate by hand but then use a calculator:

$$OR = \frac{n_{11}n_{22}}{n_{12}n_{21}} = \frac{287 \times 90}{457 \times 166} \approx \frac{25830}{75862} \approx 0.3405$$

```r
log_theta <- log(odds_ratio)
se_log_theta <- sqrt(sum(1/marginal_table))
z_critical <- qnorm(0.975) # 1.96

z_stat <- log_theta / se_log_theta
p_val <- 2 * (1 - pnorm(abs(z_stat)))
print(p_val)
```

```
## [1] 9.75664e-13
```

Test for independence using odds ratio assumes that log of the odds ratio is Normally distributed. Seeing that our p value is so low, less than $\alpha = 0.05$, we reject the null hypothesis. We believe that X and Y are not independent.

b.)

> Suppose instead that you want to determine whether the relationship between county type and Trump support is independent of state, i.e., joint independence. Write the log-linear model that would be used to assess this.

We want to see if they are independent of state. Our model would be:

$$\ln(\mu_{ijk}) = \lambda + \lambda_i^S + \lambda_j^C + \lambda_k^T + \lambda_{jk}^{CT}$$

Where state (S), county (C), and Trump support (T).

```r
model_joint <- glm(Freq ~ County * Support + State,  family = poisson(), data = poll_data)
# Check the Goodness-of-Fit
summary(model_joint)
```

```
##
## Call:
## glm(formula = Freq ~ County * Support + State, family = poisson(),
##     data = poll_data)
##
## Coefficients:
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)             2.890e+00  1.229e-01  23.513  < 2e-16 ***
## CountyUrban             1.625e+00  1.153e-01  14.090  < 2e-16 ***
## SupportYes              6.122e-01  1.309e-01   4.677 2.92e-06 ***
## State2                 -1.113e-13  1.000e-01   0.000        1
## State3                 -1.118e-13  1.000e-01   0.000        1
## State4                 -3.048e-14  1.000e-01   0.000        1
## State5                 -1.077e-13  1.000e-01   0.000        1
## CountyUrban:SupportYes -1.077e+00  1.510e-01  -7.134 9.76e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
```

```
##     Null deviance: 362.662  on 19  degrees of freedom
## Residual deviance:  51.935  on 12  degrees of freedom
## AIC: 178.97
##
## Number of Fisher Scoring iterations: 4
```

```r
pchisq(51.935, 12, lower.tail=FALSE)
```

```
## [1] 6.36628e-07
```

Using residual deviance goodness of fit test, we see that we get a low p-value, less than 0.05. This implies that we reject our null hypothesis that the model fits the data. This means our assumption of "county type and Trump support is independent of state" gives us a model that does not fit the data, implying that this is likely not the correct relationship.

c.)

> What is the predicted count under the model in part b) for State 1, urban county, and supports reelection?

```r
poll_data$predicted_counts <- fitted(model_joint)
print(poll_data)
```

```
##    State County Support Freq predicted_counts
## 1      1  Urban     Yes   53             57.4
## 2      1  Rural     Yes   17             33.2
## 3      1  Urban      No  115             91.4
## 4      1  Rural      No   15             18.0
## 5      2  Urban     Yes   71             57.4
## 6      2  Rural     Yes   33             33.2
## 7      2  Urban      No   75             91.4
## 8      2  Rural      No   21             18.0
## 9      3  Urban     Yes   58             57.4
## 10     3  Rural     Yes   40             33.2
## 11     3  Urban      No   81             91.4
## 12     3  Rural      No   21             18.0
## 13     4  Urban     Yes   34             57.4
## 14     4  Rural     Yes   34             33.2
## 15     4  Urban      No  111             91.4
## 16     4  Rural      No   21             18.0
## 17     5  Urban     Yes   71             57.4
## 18     5  Rural     Yes   42             33.2
## 19     5  Urban      No   75             91.4
## 20     5  Rural      No   12             18.0
```

We see that 57.4 is the predicted count.

d.)

> The following output describes the best fitting log-linear model among all those possible. Name this relationship model and describe what this estimated model tells you in regards to the odds ratio between county type and Trump support across states.

This model is a log linear model that includes all two way interactions but no three way interaction. This is called the uniform association or homogeneous association model.

We have coefficient estimate -1.072306 for supportYes:countyUrban. The model is log scale, so the log odds ratio is -1.072 and we convert to odds ratio

$$\hat{\theta}_{TC} = e^{-1.072306} \approx 0.342$$

. No state interaction so, for every state, the odds of urban resident supporting reelection of Trump is 0.342 times that of the odds of rural resident supporting reelection.

e.)

> Based on the model in d), write out the best fitting logistic model for Trump support as the response variable and the other two variables as potential predictors.

We see that we care about associations between state:Trump, Trump:county, county:state and some first level interactions.

If we were to now pivot to logit binomial model where the response the probability of Trump support, we care about which state, i = 2,3,4,5, and whether it is urban or rural:

$$\text{logit}(P(T = Yes)) = \alpha + \beta_{State_i} + \beta_{county}$$

α is our intercept that represents baseline of state 1 and rural.

We can get these values from the output:

- Using supportYes baseline $\alpha = 0.2666$
- Using supportYes:countyUrban: $\beta_{county} = -1.0723$
- Using supportYes:state[i]: $\beta_2^S = 0.6098$, $\beta_3^S = 0.4447$, $\beta_4^S = -0.1823$, $\beta_5^S = 0.8007$

and

$$\text{logit}(P(T = Yes)) = 0.2666 + 0.6098(\text{State2}) + 0.4447(\text{State3}) - 0.1823(\text{State4}) + 0.8007(\text{State5}) - 1.0723(\text{Urban})$$

Where we have indicator variables (State2) etc.

Problem 4: Faraway Chapter 6 Exercise #5

The dataset sexfun comes from a questionnaire from 91 couples in the Tucson, Arizona, area. Subjects answered the question "Sex is fun for me and my partner." The possible answers were "never or occasionally," "fairly often," "very often" and "almost always."

a.)

Check for symmetry, quasi-symmetry, marginal homogeneity and quasi-independence.

```r
data(sexfun)
```

```r
# Change levels from (never, fairly, very, always) to (1,2,3,4)
husb <- as.numeric(sexfun$husband)
wife <- as.numeric(sexfun$wife)

# 1. Create 'symfac' for Symmetry/Quasi-Symmetry
symfac <- factor(pmin(husb, wife) * 10 + pmax(husb, wife))

# 2. Create 'diag' for Quasi-Independence
diag <- ifelse(husb == wife, husb, 0)
diag <- factor(diag)

# Symmetry Model
model_symmetry <- glm(y ~ symfac, family = poisson, data = sexfun)
summary(model_symmetry)
```

```
##
## Call:
## glm(formula = y ~ symfac, family = poisson, data = sexfun)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.94591    0.37796   5.148 2.63e-07 ***
## symfac12    -0.44183    0.50395  -0.877   0.3806
## symfac13    -1.54045    0.69007  -2.232   0.0256 *
## symfac14    -1.02962    0.58554  -1.758   0.0787 .
## symfac22     0.13353    0.51755   0.258   0.7964
## symfac23    -0.55962    0.51755  -1.081   0.2796
## symfac24     0.06899    0.45774   0.151   0.8802
## symfac33    -0.55962    0.62678  -0.893   0.3719
## symfac34     0.25131    0.44544   0.564   0.5726
## symfac44     0.69315    0.46291   1.497   0.1343
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 33.5846  on 15  degrees of freedom
## Residual deviance:  4.0552  on  6  degrees of freedom
## AIC: 78.521
##
## Number of Fisher Scoring iterations: 4
```

```r
pchisq(deviance(model_symmetry), df.residual(model_symmetry), lower.tail=FALSE)
```

```
## [1] 0.6692131
```

```r
# Quasi-Symmetry Model
model_qs <- glm(y ~ husband + wife + symfac, family = poisson, data = sexfun)
pchisq(deviance(model_qs), df.residual(model_qs), lower.tail=FALSE)
```

```
## [1] 0.9471722
```

```r
# Marginal Homogeneity (Difference between Sym and QS)
anova(model_symmetry, model_qs, test="Chi")
```

```
## Analysis of Deviance Table
##
## Model 1: y ~ symfac
## Model 2: y ~ husband + wife + symfac
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         6     4.0552
## 2         3     0.3660  3   3.6891    0.297
```

Both symmetry and quasi-symmetry models hold. We find that a high p-value in our ANOVA test between the models so it is the case that marginal homogeneity holds.

```r
# Quasi-Independence Model (the diagonal is counted as separate, but the rest is marginal)
model_qi <- glm(y ~ husband + wife + diag, family = poisson, data = sexfun)
summary(model_qi)
```

```
##
## Call:
## glm(formula = y ~ husband + wife + diag, family = poisson, data = sexfun)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.04364    0.55571  -0.079  0.93741
## husbandfairly  0.30474    0.42199   0.722  0.47020
## husbandvery    0.40623    0.39685   1.024  0.30601
## husbandalways  0.86104    0.39279   2.192  0.02837 *
## wifefairly     1.45694    0.50999   2.857  0.00428 **
## wifevery       1.13049    0.53063   2.130  0.03313 *
## wifealways     1.63808    0.52432   3.124  0.00178 **
## diag1          1.98955    0.67207   2.960  0.00307 **
## diag2          0.36140    0.54849   0.659  0.50996
## diag3         -0.10678    0.65244  -0.164  0.87000
## diag4          0.18357    0.47778   0.384  0.70081
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 33.5846  on 15  degrees of freedom
```

```
## Residual deviance:  5.1158  on  5  degrees of freedom
## AIC: 81.582
##
## Number of Fisher Scoring iterations: 4
```

```r
pchisq(deviance(model_qi), df.residual(model_qi), lower.tail=FALSE)
```

```
## [1] 0.4019123
```

The p-value is high so this suggests a good fit.

Overall, because we have marginal homogeneity, all 3 models can fit the data because the marginals are good at explaining the data.

b.)

> Develop a score-based model. Find some good-fitting scores.

```r
library(coin)
```

```
## Loading required package: survival

##
## Attaching package: 'survival'

## The following objects are masked from 'package:faraway':
##
##     rats, solder

##
## Attaching package: 'coin'

## The following object is masked from 'package:faraway':
##
##     vision
```

```r
sf_t = xtabs(y ~ husband + wife, sexfun)
sexfun$husband <- ordered(sexfun$husband)
sexfun$wife <- ordered(sexfun$wife)
lbl_test(as.table(sf_t))
```

```
##
##  Asymptotic Linear-by-Linear Association Test
##
## data:  wife (ordered) by husband (never < fairly < very < always)
## Z = 3.1645, p-value = 0.001553
## alternative hypothesis: two.sided
```

```r
test_scores <- independence_test(y ~ husband + wife,
                                 data = sexfun,
                                 weights = ~y,
                                 scores = list(husband = 1:4,
                                               wife = 1:4))

print(test_scores)
```

```
##
##  Asymptotic General Independence Test
##
## data:  y by husband(ordered), wife(ordered)
## maxT = 5.0248, p-value = 1.008e-06
## alternative hypothesis: two.sided
```

Scored model tells us there is strong evidence against independence.

We can test some other scores:

```r
independence_test(y ~ husband + wife, data = sexfun, weights = ~y, scores = list(husband = exp(1:4), wi
```

```
##
##  Asymptotic General Independence Test
##
## data:  y by husband(ordered), wife(ordered)
## maxT = 5.6082, p-value = 4.088e-08
## alternative hypothesis: two.sided
```

```r
independence_test(y ~ husband + wife, data = sexfun, weights = ~y, scores = list(husband = log(1:4), wi
```

```
##
##  Asymptotic General Independence Test
##
## data:  y by husband(ordered), wife(ordered)
## maxT = 4.517, p-value = 1.254e-05
## alternative hypothesis: two.sided
```

They all tell us the same thing, that there is strong evidence against independence.

Problem 5

The 2006 General Social Survey asked subjects, "Would you say that astrology is very scientific, sort of scientific, or not at all scientific?" The table below cross-classifies their responses with their highest degree. Our goal is to investigate whether educated people put less credence in astrology. View the opinion about astrology as the response variable and education level as a quantitative predictor variable with scores ui=0, 1, 2, 3 and 4.

a.)

State the proportional model and its assumptions, and interpret the parameters.

3 outputs of not at all scientific, sort of, and very. The model tells us that we believe that the effect of the predictor is the same for all x (use same Beta). We assume that observations are independent. We assume linearity of relationship between scores and logit of the culmulative probabilities. We assume the "astrology is scientific" is ordered response.

We can model that proportion with logit:

$$\text{logit}[P(Y \leq j|x)] = \log\left(\frac{P(Y \leq j|x)}{1 - P(Y \leq j|x)}\right) = \alpha_j - \beta x, \quad j = 1, \ldots, J-1$$

Where x is the quantitative predictor of education levels and j is the scores for 1 (not at all), 2 (sort of), and 3 (very) scientific levels. We have a comparison level so we can actually ignore j = 3.

Two models are one for not at all ($\text{logit}[P(Y \leq 1|x)] = \alpha_1 - \beta x$) and one for not at all + sort of ($\text{logit}[P(Y \leq 2|x)] = \alpha_2 - \beta x$)

For the parameters:

- α_j tells us the log-odds of moving into category j or less when education level is at $x = 0$.
- slope β coefficient tell us effect of education on perception of astrology. Same for all levels x, but perhaps increased in intensity because it is multiplication.

b.)

Based on the output below, calculate the estimated cumulative odds ratio comparing the lowest level of education '< High school' with the highest level 'Graduate', and interpret your conclusions.

We simply calculate the odds ratio for 4 levels change given the coefficent Beta that is provided:

$$OR_{0 \text{ vs } 4} = e^{-(4\times(-0.4614))} = e^{1.8456} \approx 6.33$$

Note that odds ratio is negative because it tells us probability of going down, so its opposite for going up in the groups.

We find that, in regards to responding to astrology as "not at all scientific", someone who is graduate degree is 6.33 times higher odds of answering "not at all scientific" compared to someone who did not finish high school.

c.)

Calculate the estimates cell counts for 'Graduate' education.

$x = 4$, total count $n = 148 + 23 + 1 = 172$, intercept $\alpha_1 = 0.0264$ (for "Not at all") and $\alpha_2 = 2.3137$ (for "Sort of"), slope -0.4613754

Our model is: $\text{logit}[P(Y \leq j)] = \alpha_j - \beta x$

- $L_1 = 0.0264 - (-0.4613754 \times 4) = 0.0264 + 1.8455 = 1.8719$
- $L_2 = 2.3137 - (-0.4613754 \times 4) = 2.3137 + 1.8455 = 4.1592$

Take inverse logit:

- $P(Y \leq 1) = \frac{e^{1.8719}}{1+e^{1.8719}} \approx 0.8667$
- $P(Y \leq 2) = \frac{e^{4.1592}}{1+e^{4.1592}} \approx 0.9846$

Calculate final probabilities:

- $\pi_1(\text{Not at all}) = P(Y \leq 1) = 0.8667$
- $\pi_2(\text{Sort of}) = P(Y \leq 2) - P(Y \leq 1) = 0.9846 - 0.8667 = 0.1179$
- $\pi_3(\text{Very}) = 1 - P(Y \leq 2) = 1 - 0.9846 = 0.0154$

Compute cell counts by multiplying by $n = 172$ to find that not at all predicts 149.07, sort of is 20.29, and very is 2.65.